

S.No. : 63

NCS 4403

No. of Printed Pages : 04

Following Paper ID and Roll No. to be filled in your Answer Book.

PAPER ID : 43211

**Roll
No.**

B. Tech. Examination, 2026

(Even Semester)

SOFTWARE ENGINEERING

Time : Three Hours]

[Maximum Marks : 60

Note :- (i) Attempt all questions.

(ii) Draw neat diagrams wherever necessary.

SECTION – A

1. Attempt all parts of the following :

8×1=8

(a) Define software engineering.

(b) What is software life cycle model?

(c) Define software cost estimation.

(d) Define extreme programming (XP).

/ P. T. O.

- (e) What is design pattern?
- (f) Define integration testing.
- (g) What is SEI-CMM?
- (h) Define software project scheduling.

SECTION – B

2. Attempt any two parts of the following : $2 \times 6 = 12$

- (a) Compare waterfall, incremental and spiral models with advantages and limitations.
- (b) Explain formal requirement specification using algebraic or axiomatic method.
- (c) Describe UML diagrams used in software design and explain any two with example.
- (d) Explain COCOMO model and estimate effort for a project of 30 KLOC. (Assume suitable constants.)

SECTION – C

Note :- Attempt all questions. Attempt any two parts from each questions. $8 \times 5 = 40$

- 3. (a) Explain phases of software development life cycle (SDLC) with neat diagram.
- (b) Discuss software cost estimation techniques and staffing level estimation.
- (c) Explain software maintenance types with real-life examples.
- 4. (a) Explain SRS document structure and its characteristics.
- (b) Discuss requirement verification and validation techniques.
- (c) Explain CASE tools and their role in software development.
- 5. (a) Compare function-oriented design and object-oriented design.
- (b) Explain unit-testing, integration testing and system testing strategies.
- (c) Describe debugging techniques and their importance.

[P. T. O.]

6. (a) ✓ Explain ISO-9001 and SEI-CMM levels.
- (b) ✓ Discuss software project planning and risk management.
- (c) Explain software reliability and fault tolerance mechanisms.
